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Learning Lean 4

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Abstract

Working notes and exercises from learning the Lean 4 theorem prover, following *An Introduction to Lean 4* by Cosme Llópez and Gong. Covers basic syntax and natural-deduction proofs, paired with a runnable Lean library.

1. Basic Syntax

1.a. Basic syntax

The runnable version of this chapter lives in `LearningLean/BasicSyntax.lean`.

1.a.i. *Comments:*

```
-- This is a single-line comment.

/-
This is a multi-line comment.
Here is another line.
-/
```

1.a.ii. *Inspecting terms and types with `#check`:*

`#check` asks Lean for the type of an expression without evaluating it.

```
#check true          -- Bool
#check 42            -- Nat
#check "Hello, World!" -- String
#check Type          -- Type 1
#check Nat           -- Type
#check ['h', 'e', 'l', 'l', 'o'] -- List Char
```

1.a.iii. *Printing definitions with `#print`:*

`#print` shows how something is defined.

```
#print Bool
```

`Type` itself cannot be printed with `#print`, because it is a type rather than a definition.

1.a.iv. *Definitions and functions:*

Published Jul 17, 2026

```
def pi : Float := 3.1415926

#check pi
```

```
-- An anonymous function (lambda) taking two natural numbers.
#check λ (a b : Nat) => a + b
```

2. Natural Deduction

2.a. Natural deduction

The Lean proofs for this chapter live in `LearningLean/NaturalDeduction.lean`, inside the `Exercises` namespace.

2.a.i. *5.5 Reduction to the absurd.* $P \Rightarrow Q,\ \neg Q \vdash \neg P$:

This is a very useful technique. Validity of $P \Rightarrow Q,\ \neg Q \vdash \neg P$ is proved with:

$$
\begin{array}{lll}
1 & P \Rightarrow Q & \\
2 & \neg Q & \\
3 & \quad \vert\ \ \underline{P} & \text{H} \\
4 & \quad \vert\ \ Q & \text{E}\Rightarrow\ 1{,}3 \\
5 & \quad \vert\ \ \neg Q & \text{IT}\ 2 \\
6 & \neg P & \text{I}\neg\ 3{,}4{,}5
\end{array}
$$

The sub-derivation (lines 3–5) discharges the hypothesis P: assuming P, [$\Rightarrow$-elimination] on lines 1 and 3 gives Q, while line 2 reiterates $\neg Q$. The contradiction lets us introduce $\neg P$ ($\neg$-introduction over lines 3, 4, 5).

In Lean this is theorem `T55`, where `¬P` is definitionally `P → False`, so the whole proof is just function composition:

```
theorem T55 (h1 : P → Q) (h2 : ¬Q) : ¬P :=
  (h2 ∘ h1)
```

2.a.ii. *Conjunction introduction.* $P,\ P \Rightarrow Q \vdash P \wedge Q$:

Theorem `T51` builds the pair `⟨proof of P, proof of Q⟩` directly, using the anonymous-constructor notation `⟨_, _⟩`:

```
theorem T51 (h1 : P) (h2 : P → Q) : P ∧ Q :=
  ⟨h1, h2 h1⟩
```